Causal Attention Does Not Necessarily Mean Causal LM

Part II: Other Objectives and Examples

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1 The distinction

A model can use *causal attention* without being a *causal language model*. Causal attention is an information-flow restriction; causal language modeling is a particular learning objective.

Causal attention	**Causal language modeling**
Restricts a position from attending to future positions.	Trains the model to predict the next token from preceding tokens.
Describes how information flows.	Describes what prediction task is learned.
Does not require token prediction.	Uses an autoregressive next-token objective.
Can support classification, regression, forecasting, etc.	Predicts tokens in a vocabulary.

2 Example 1: Causal Transformer for sequence classification

Suppose a model receives sensor readings

$$10,\ 12,\ 15,\ 13,\ 18$$

and must classify the sequence as *normal* or *anomalous*. A Transformer can use a causal mask so that each time-step representation uses only the current and preceding readings. A classification head then predicts a class:

$$\hat{y} = \text{softmax}(Wh + b),$$

where h is the representation supplied to the classifier. The model may be trained using cross-entropy:

$$\mathcal{L}_{\text{class}} = -\sum_{c} y_c \log \hat{y}_c.$$

Its attention is causal, but its target is a class label, not the next token. It is therefore a causal-attention model, not a causal LM merely by virtue of its mask.

3 Example 2: Causal attention for time-series forecasting

Suppose a model receives hourly temperature readings $T_1, \ldots, T_t$ and predicts tomorrow's temperature. At time t, only measurements available up to time t should be used. Future measurements such as T_{t+1} must not leak into the prediction. Causal attention enforces this restriction.

The model predicts a numerical value:

$$\widehat{T}_{t+1} = f_\theta(T_1, \ldots, T_t),$$

and may be trained with mean squared error:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (\widehat{T}_i - T_i)^2.$$

The output is a real-valued temperature, not a token from a language vocabulary. Hence this is causal attention without causal language modeling.

4 Other tasks that can use causal attention

The same idea applies to sequence classification, numerical regression, time-series forecasting, and online decision-making. In each case, the mask prevents access to future positions, while a task-specific loss determines what the model learns to predict.

5 Can a causal LM be fine-tuned for classification?

Yes. A model may first be pretrained using causal next-token prediction and later be fine-tuned for a downstream task:

$$\text{causal-LM pretraining} \quad \longrightarrow \quad \text{classification fine-tuning.}$$

During pretraining, it learns next-token prediction. During classification fine-tuning, a classification head may be trained to predict labels.

It is useful to distinguish the model's pretraining history from its current objective:

- It is accurate to say that the model was *pretrained as a causal LM*.
- If the current fine-tuning loss is only classification, that stage is not itself causal language modeling.
- The model may retain causal attention even while being trained for a non-language task.

Using a GPT-like architecture for classification does not mean the classification objective is next-token prediction.

6 A practical test: ask two questions

Question 1: What can each position attend to?

If position t can attend only to positions $1, \ldots, t$, the model uses causal attention. If it can also attend to later positions, its attention is bidirectional or otherwise non-causal.

Question 2: What is the model trained to predict?

If it is trained to predict

$$P(x_t \mid x_1, \ldots, x_{t-1}),$$

it is performing causal language modeling. If it is trained to predict a class, a real number, an action, or another target, that objective is not causal language modeling, even if the attention is causal.

7 Summary

Model/task	**Attention**	**Objective**
GPT-style CLM pretraining	Causal	Next-token prediction
Standard BERT pretraining	Bidirectional	Masked-token prediction
Causal sensor classifier	Causal	Class prediction
Causal time-series forecaster	Causal	Numerical forecasting
GPT fine-tuned for classification	Typically causal	Class prediction during fine-tuning

8 Final takeaway

Causal attention : restricts information flow to past and present,
Causal LM : learns autoregressive next-token prediction.

Therefore, a causal-attention model can perform classification, regression, or time-series forecasting without being a causal LM. The mask tells us what information the model can use; the objective tells us what task it is learning.